

Worksheet: Kettle Boiler

Name(s) _____

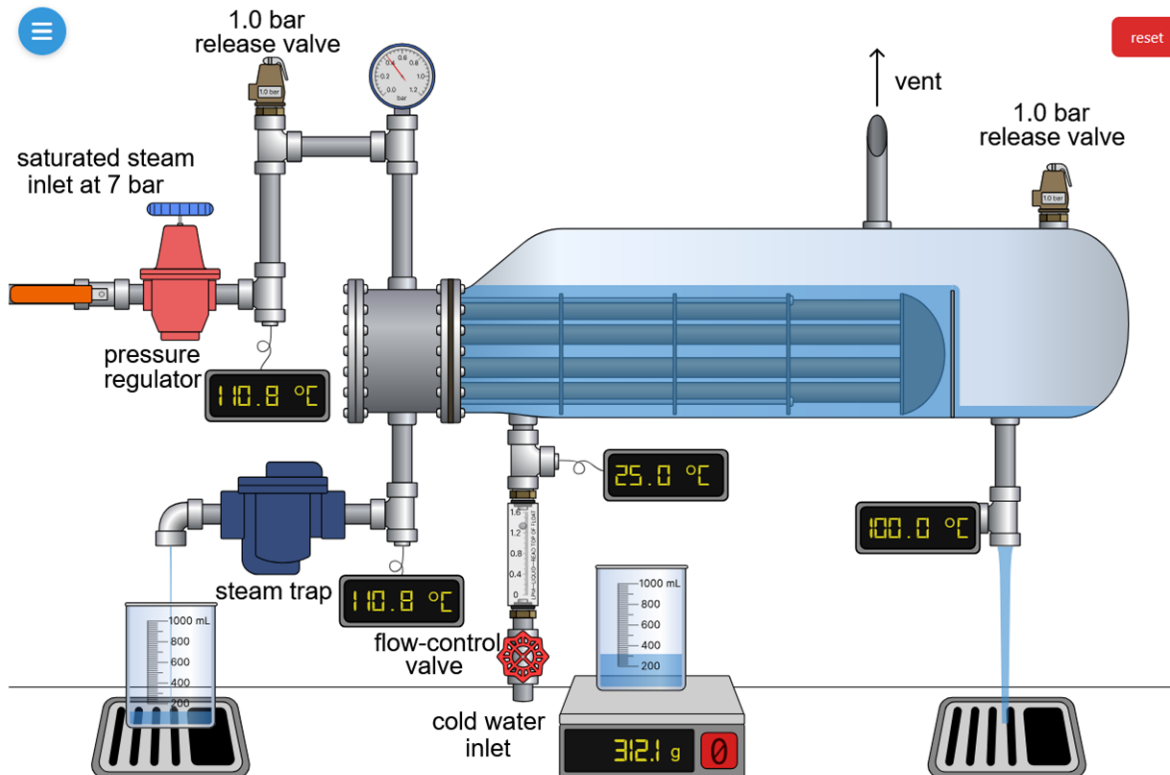
A kettle boiler is a type of heat exchanger used to vaporize liquids. They are commonly used as reboilers at the bottom of industrial distillation columns. In a distillation application, liquid enters the shell side of the boiler from the stripping section. Part of this liquid vaporizes and is sent back up the column; the remainder flows out as the bottom product. This experiment uses a small-scale kettle boiler where steam is a heat source to boil water. The important principles for this type of heat exchanger include boiling and condensing heat transfer. This experiment measures the heat transfer rate, which is compared to predictions based on typical boiling and condensing heat transfer coefficients.

Student learning objectives

After completing this activity, you will be able to:

1. control the liquid-to-vapor ratio leaving the boiler by adjusting process parameters.
2. measure the heat duty of a boiler using a tube-side energy balance.
3. measure the overall heat transfer coefficient for a heat exchanger that involves phase change on both sides.
4. predict the heat transfer rate for this type of heat exchanger from industrial correlations.

Equipment



Kettle Boiler Specifications

Tube outside diameter (D_o) = 5/8 inch

Tube inside diameter (D_i) = 1/2 inch

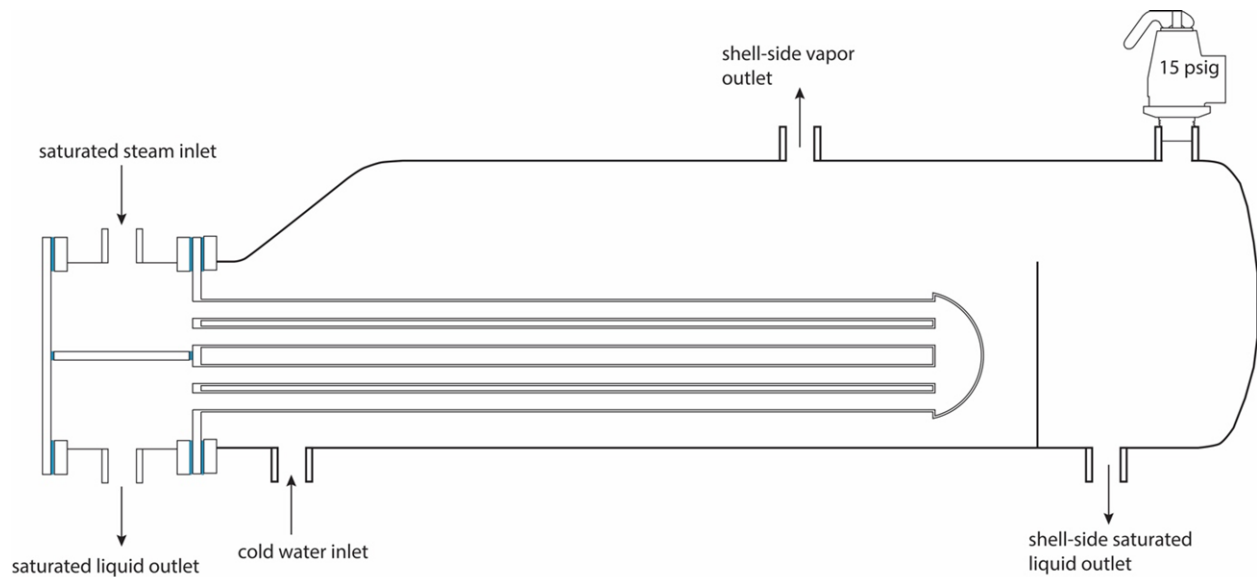
Tube length (L) = 21.6 inches

Tube material: copper

Geometry: 4 tubes per pass, 2 tube passes, 8 tubes total

Questions to answer before starting experiment

1. Examine the following schematic diagram of the kettle boiler used in this lab. Make sure you understand the flow path for steam on the tube side and for liquid water and water vapor on the shell side. You can sketch the flow paths on this diagram.



2. For the relatively low cold-water flow rates entering the shell in this experiment, the shell-side liquid undergoes pool boiling and is vented to the atmosphere. The steam entering the tube side is at a pressure above atmospheric and condenses on the inner walls of the tubes. The tube-side steam pressure can be adjusted with the pressure regulator seen in the schematic on page 1. What do you expect to be the effect of increasing the steam pressure on the rate of heat transfer? Explain your answer.

3. Consider a control volume that encompasses the tube side of the boiler, from the steam inlet to the condensate outlet. Consider how an enthalpy balance on this control volume could be used to determine the rate of heat transfer from the tube side to the shell side of the exchanger. What needs to be measured?

4. What do you expect to happen to the ratio of vapor to liquid leaving the shell side if the heat transfer rate is increased?

Experimental Procedure

1. For the first trial, use a steam gauge pressure of 0.20 bar and a cold-water rate of 0.40 L/min.
2. Open the red valve on the cold-water inlet and adjust it with the mouse until the rotameter reads the desired cold-water flow rate in L/min. Record the water inlet temperature from the thermocouple in the inlet line. Note that the boiler shell will begin to fill with water.
3. Open the steam shutoff valve by clicking on the red handle. The valve is open when the handle is parallel to the pipe.
4. Click and slowly drag on the blue valve handle on the steam pressure regulator and watch the pressure gauge downstream of the regulator. Note that tightening the valve handle increases the pressure. Adjust to the desired pressure, but do not exceed 1 bar gauge pressure, as the pressure relief valve opens at that pressure.
5. Visually monitor the liquid level in the boiler shell and wait for it to stabilize.
6. Measure the flow rate of the hot water that overflows the weir and drains from the shell of the boiler through the exit pipe on the right. Place the beaker on the scale and click the "zero" button to tare the scale. Then move the beaker under the stream and start a timer. After collecting at least 200 mL of water, stop the timer and move the beaker to the scale. In Table 1, record the time and mass of water collected. Also, record the temperature of this stream from the outlet pipe thermocouple.
7. Measure the temperature and flow rate of the condensed steam leaving the steam trap following the same procedure used for the shell-side exit stream and record in Table 1.
8. For trial 2, keep the cold-water flow rate the same (0.40 L/min) and increase the steam pressure significantly, but below the maximum of 1 bar, following the same procedures. Record the measurements in Table 2.

9. For trial 3, adjust the steam pressure to 0.20 bar and increase the cold-water flow rate to a significantly larger value within the rotameter range. Follow the same procedures and record the measurements in Table 3.

Table 1

Tube side		Shell side	
Steam inlet pressure (bar)	0.20	Water inlet flow rate (L/min)	0.40
Temperature of inlet steam (°C)		Water inlet temperature (°C)	
Mass of steam condensate collected (g)		Mass of outlet stream collected (g)	
Time of collection (s)		Time of collection (s)	
Mass flow rate of steam (g/s)		Mass flow rate for liquid water outlet (g/s)	
Temperature of steam condensate(°C)		Temperature of liquid water outlet (°C)	

Table 2

Tube side		Shell side	
Steam inlet pressure (bar)		Water inlet flow rate (L/min)	0.40
Temperature of inlet steam (°C)		Water inlet temperature (°C)	
Mass of steam condensate collected (g)		Mass of outlet stream collected (g)	
Time of collection (s)		Time of collection (s)	
Mass flow rate of steam (g/s)		Mass flow rate for liquid water outlet (g/s)	
Temperature of steam condensate(°C)		Temperature of liquid water outlet (°C)	

Table 3

Tube side		Shell side	
Steam inlet pressure (bar)	0.20	Water inlet flow rate (L/min)	
Temperature of inlet steam (°C)		Water inlet temperature (°C)	
Mass of steam condensate collected (g)		Mass of outlet stream collected (g)	
Time of collection (s)		Time of collection (s)	
Mass flow rate of steam (g/s)		Mass flow rate for liquid water outlet (g/s)	
Temperature of steam condensate(°C)		Temperature of liquid water outlet (°C)	

Analysis

Use the following notation in developing the equations for data analysis.

Enthalpies (J/kg)

H_V : saturated steam entering the tube side

H_L : saturated liquid condensate leaving the tube side

Mass flow rates (kg/s)

$\dot{m}_s$: stream entering the tube side

$\dot{m}_c$: cold water entering the shell

$\dot{m}_V$: saturated vapor leaving the shell

$\dot{m}_L$: saturated liquid leaving the shell

Heat transfer rate (W)

q : heat transfer rate

5. Write an expression for q in terms of the appropriate enthalpies and measured quantities for the tube side.

6. Sketch a simple flow diagram and write the mass balance for a control volume that encompasses the shell side of the heat exchanger. Derive an expression to predict the mass flow rate of vapor leaving the shell.

Homework questions

1. Compute the heat transfer rate for each experiment using the tube-side enthalpy balance. What are the effects of increasing the steam pressure and of increasing the shell-side liquid flow rate?
2. Use the heat transfer rates from question 1 to compute the boiling heat flux (W/m^2) for each trial based on the total outside tube area.
3. Compute the overall heat transfer coefficient from your measurements for all three trials.
4. Compute the critical heat flux for boiling on the outside of a horizontal tube under the experimental conditions from the following equation:

$$\left(\frac{q}{A_o}\right)_{max} = 0.12\lambda\rho_v^{1/2}[\sigma g(\rho_L - \rho_v)]^{1/4}$$

where λ is the enthalpy of vaporization, σ the surface tension, ρ_v and ρ_L the vapor and liquid density. Use the ideal gas law to predict the vapor density under these conditions, assuming a surface tension of $58.91 \times 10^{-3} \text{ N/m}$ at 100°C . How do the heat fluxes for your trials compare to the critical flux? Does this make sense?

5. Determine the fraction of the entering shell-side liquid that vaporizes using a mass balance on the shell for all three trials. How does the vaporized fraction depend on tube-side steam pressure and shell-side liquid feed rate?
6. Predict the overall heat transfer coefficients and heat transfer rates for trials 1 and 2 using the heat transfer coefficient equations and procedures presented in the attached notes. Note that this model does not account for changing the cold-water feed to the shell side.

Prediction of the heat transfer rate

The rate of heat transfer for the kettle boiler can be predicted using an overall heat transfer coefficient from

$$q = U_o A_o \Delta T$$

where ΔT is the mean temperature difference between the hot and cold sides of the exchanger and A_o is the total outside surface area of the boiler tubes. Because phase change is taking place on both sides of the exchanger, ΔT is the difference in the saturation temperatures at the corresponding tube- and shell-side pressures.

$$\Delta T = T_{sat,tube} - T_{sat,shell}$$

Neglecting fouling, the overall heat transfer coefficient in terms of the inner and outer film coefficients is:

$$U_o = \frac{1}{\frac{D_o}{D_i h_i} + \frac{D_o x_w}{\bar{D}_L k_w} + \frac{1}{h_o}}$$

where x_w and k_w are the thickness and thermal conductivity of the tube wall, and $\bar{D}_L$ is the log-mean diameter of the tube. The boiling heat transfer coefficient for nucleate boiling from a horizontal tube can be found from the following Nusselt number correlation (Cornwell and Houston, 1994¹).

$$Nu = \frac{h_o D_o}{k} = 9.7 p_c^{0.5} F_p Re_b^{0.67} Pr^{0.4}$$

where p_c is the critical pressure of the liquid in bar, and the boiling Reynolds number is defined as

$$Re_b = \frac{(q/A_o) D_o}{\mu \lambda}$$

where μ is the liquid viscosity and λ the enthalpy of vaporization at the shell-side saturation temperature. The Prandtl number is:

$$Pr = \frac{c_p \mu}{k}$$

where c_p , μ , and k are the specific heat, viscosity, and thermal conductivity of the liquid. Table 4 should be used to evaluate fluid properties. A factor to account for the effect of pressure is:

$$F_p = 1.8 p_r^{0.17} + 4 p_r^{1.2} + 10 p_r^{10}$$

where p_r is the reduced pressure, defined as

$$p_r = p/p_c$$

The critical pressure for water is 220.6 bar. An expression for the condensation heat transfer coefficient in horizontal tubes is given by²

$$h_i = 0.763 k \left[\frac{\rho^2 g}{\mu \left(\frac{\dot{m}}{N_T L} \right)} \right]^{1/3}$$

where ρ , μ , and k are the density, viscosity, and thermal conductivity of the condensed liquid at the saturation temperature, N_T is the number of tubes in the boiler, and L the length of an individual tube. The rate of condensation per length of tubing that appears in the expression above can be rewritten in terms of the heat flux as follows:

$$\left(\frac{\dot{m}}{N_T L} \right) = \left(\frac{q}{A_o} \right) \left(\frac{\pi D_o}{\lambda} \right)$$

where λ is the heat of vaporization (J/kg). It can thus be seen that an increased heat flux reduces the inner heat transfer coefficient due to a thicker condensate layer on the tube wall.

The prediction of the heat transfer rate is complicated by the fact that U_o depends on the heat flux, since both h_o and h_i depend on the flux. An iterative process must be used to solve the problem:

¹ K. Cornwell and S. D. Houston, "Nucleate pool boiling on horizontal tubes: a convection-based correlation," Int. J. Heat Mass Transfer, Vol. 37, Suppl. 1, pp. 303-309, (1994).

² R. Smith, "Chemical Process Design and Integration," Wiley, 2005 (p. 338).

an initial estimate of the overall heat transfer coefficient is used to predict the heat flux, which in turn is used to compute a better estimate of the overall heat transfer coefficient, and so on.

$$\frac{q}{A_o} = U_{o,est} \Delta T$$

The initial estimate can be a value within the typical range of U_o for a water boiler using condensing steam as the heat source³, which is 1400 – 2300 W/m²·K.

Table 4. Properties of saturated water and steam⁴ as a function of the gauge pressure relative to standard atmospheric pressure.

$p(\text{bar } g)$	T_{sat} (°C)	ρ_L $\left(\frac{kg}{m^3}\right)$	c_{pL} $\left(\frac{J}{kg \cdot K}\right)$	k_L $\left(\frac{W}{m \cdot K}\right)$	$\mu_L \times 10^4$ (Pa · s)	H_L $\left(\frac{kJ}{kg}\right)$	H_V $\left(\frac{kJ}{kg}\right)$
0	100.0	959.1	4218	0.6794	2.822	419.1	2676
0.1	102.7	957.3	4221	0.6804	2.746	430.3	2680
0.2	105.1	955.5	4224	0.6813	2.679	440.8	2684
0.3	107.5	953.8	4228	0.6821	2.619	450.6	2687
0.4	109.6	952.2	4231	0.6828	2.565	459.8	2690
0.5	111.7	950.7	4233	0.6833	2.515	468.4	2693
0.6	113.6	949.2	4236	0.6838	2.470	476.7	2696
0.7	115.5	947.8	4239	0.6842	2.428	484.5	2699
0.8	117.2	946.5	4242	0.6846	2.390	492.0	2702
0.9	118.9	945.1	4245	0.6849	2.354	499.1	2704
1.0	120.5	943.9	4247	0.6852	2.321	505.9	2706

³ R. H. Perry, D. W. Green, J. O. Maloney, “Perry’s Chemical Engineers’ Handbook, 6th edition,” McGraw-Hill (1984).

⁴ A. Crabtree and M. Simon-Tov, “Thermophysical Properties of Saturated Light and Heavy Water for Advanced Neutron Source Applications (ORNL/TM-12322),” Oak Ridge National Laboratory (1993).